Examining learnability of OT grammar: A view from Kashmiri loanwords

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1. Purpose of the study

Loanword phonology has been a focus of phonological research for a long time. From its interaction with perception (Yip 1993, Silverman 1991) to different levels of transmission in language acquisition, loanword phonology has gone through significant development. The aim of this presentation is-

- **to propose a set of OT constraints to account for the phonotactics (*specifically aspiration and epenthesis*) of Kashmiri loanword phonology, and**
- **to check the learnability of the proposed grammar following Tesar and Smolensky (1995)'s learning algorithm.**

2. Overview: Kashmiri loanword phonology

Just like the other Indo-Aryan languages, Kashmiri borrowed largely from Sanskrit, Hindi-Urdu, Persian, Perso-Arabic, and recently from English. Koul(2005) however expressed doubt about whether Arabic borrowings took place only through Persian or directly. In the following section, we will discuss the Persian (including Perso-Arabic), Hindi-Urdu, and English lexical borrowings in light of their phonological alternations. The main focus is on -

- **Aspiration of voiceless stops at coda position.**

Persian	Kashmiri	Gloss	Hindi-Urdu	Kashmiri	Gloss	English	Kashmiri
/paːk/	/paːkh/	*pure*	/mulk/	/mulukh/	*country*	doctor	/daːkhṭar/
/caːlaːk/	/caːlaːkh/	*clever*	/paːp/	/paːph/	*sin*	minute	/minaṭh/
/naːzuk/	/noːzukh/	*delicate*	/raːt/	/raːth/	*night*	rate	/reːṭh/
/poːšaːk/	/poːšaːkh/	*dress*	/davaːt/	/davaːth/	*inkpot*	paper	/peːpar/

Table 1. Partial list of loanwords in Kashmiri (Koul 2005)

- **Insertion of vowel /ɨ/ in cosonant clusters.**

English	Kashmiri	Gloss
/eɜplem/	/pleːn/	*aeroplane*
/klɜːk/	/klərik/	*clerk*
/skuːtə/	/sɨkuːṭar/	*scooter*
/steɪ.ʃən/	/sɨṭesaːn/	*station*
/spiːd/	/sɨpiːd/	*speed*
/skuːl/	/sɨkuːl/	*school*
/nɜːs/	/narəs/	*nurse*
/lɪp.stɪk/	/lip.sɨ.tik/	*lipstick*

Table 2. Partial list of loanwords in Kashmiri (Koul 2005)

3. OT Analysis

General Constraints:

- Onset and Coda are optional in Kashmiri, and the native grammar allows consonant clusters in all three positions.
- Thus, well-formedness constraints **ONSET**(a syllable must have onset), **NO CODA**(syllables are open) (Ito 1989; Prince & Smolensky 1993) and ***COMPLEX**(complex onset and coda are not allowed) are dominated.
- Koul (2005) explained that the word-initial consonant clusters in Kashmiri are grammatical only if a stop/affricate/fricative is followed by /r/ while word-final clusters are acceptable only if a nasal/fricative is followed by a stop.
- However, the existence of loanwords like /klaːb/ *'club'* or /pleːn/ *'aeroplane'*(Kachru 2006) suggests that the clusters follow **Sonority Sequencing Principle**(initial clusters should rise in sonority while the final clusters should fall). We use **SONORITY** as a markedness constraint to account for this.

OT Analysis(contd.)

- Therefore, constraint ranking: **SONORITY » FAITHFULNESS » *COMPLEX,NO-CODA,ONSET**

Input:/*drog*/	SONORITY	IDENT-IO	*COMPLEX	NO-CODA	ONSET
➪ a.**drog**			*	*	*
b. dro		!*	*		*
c. dirog		!*		*	*

Table 3. General syllable structure in Kashmiri

Case 1: Aspiration of Stops

- Analysing the data from Table 1, it is clear that voiceless unaspirated plosives (/k/,/p/,/t/) are aspirated (/kh/,/ph/,/th/) in syllable-final (/paːkh/,/paːph/,/minaṭh/ etc.) position. The constraint to account for this instance- ***T[-vce,-sg]syll**(voicelss unaspirated plosives are not allowed syllable-finally).
- ***[+spread glottis]** is a context-free markedness constraint that disallows aspiration, and **IDENT-IO** is a faithfulness constraint that correspondents in input and output have identical features.
- Since aspiration of voiceless stops is a marked feature of Kashmiri, it is ranked as the highest constraint that violates the faithfulness constraint IDENT-IO as well as *[+spread glottis] yet outputs a grammatical element.
- Constraint ranking: ***T[-vce,-sg]syll » IDENT-IO »*[+spread glottis]**

Input:/*caːlaːk*/	*T[-vce,-sg]syll	IDENT-IO	*[+spread glottis]
a. caː.laːk	*!		
➪ b. **caː.laːkh**		*	*

Table 4. Aspiration in Kashmiri loanwords

Case 2: Epenthesis

- /ɨ/ is inserted when consonant clusters violate Sonority Sequencing Principle.
- Similar case in Bangla (Kar 2013; Nagarajan 2014), Korean (Choi 2002), and Punjabi (Mahmood et al 2011). Kar (2013)stressed examining the epenthesis in clusters consisting of coronal [/s/+stop](eg. /skuːl/->/iskul/).
- Repair strategy similar to Kashmiri is seen in Lenakel Kager (1999) (/t-n-ak-ol/->/ti.na.gol/)
- The general well-formedness constraints I propose are- **SONORITY, DEP-IO**(no epenthesis allowed), **CONTIGUITY** (no medial epenthesis or deletion allowed) (Kager1999; Gouskova 2002).
- Why not (iskul or skuli)?- Alignment constraints are introduced to avoid initial or final epenthesis. **ALIGN-L** restricts epenthesis on the left edge of the Prosodic word, and **ALIGN-R** restricts epenthesis on the right edge.
- Why /sɨkuːl/ and not /səkuːl/?- /ɨ/ appears to be the epenthetic vowel with features [+HIGH,-BACK,-LOW,-ROUND]. We can count these as the four features as marked in the language. According to Shademan's(2002) **DEP(MF)**, a surface instance of [+LOW], [+HIGH], [+ROUND], or [+BACK] must have an identical underlying correspondent.
- Proposed constraint ranking: **SONORITY,ALIGN-L, DEP(MF)» ALIGN- R» DEP-IO, CONTIGUITY**

Input:/spiːd/	SONORITY	ALIGN-L	DEP(MF)	ALIGN-R	DEP-IO	CONTIGUITY
a. spiːd	*!					
b. is.piːd		*!	*		*	
➪ c.sɨ.piːd				*	*	*
d. sə.piːd			!*	*	*	*
e. is.piːd	*!	*			*	*

Table 5. Epenthesis in Kashmiri loanwords

4. Checking Learnability of the constraints

- **Question:** Given a set of surface forms of the target language (Kashmiri), is it possible for the learner to discover the correct constraint ranking?
- **Assumptions:** A child only has access to the optimal candidates (positive evidence) of the grammar in their linguistic environment.
- Constraint reranking strictly adheres to the violation of constraints instead of satisfaction.
- A top-ranked constraint can be violated by an optimal candidate as long it is violated at least as many times by other outputs.
- The algorithm demotes constraints in winner-marks immediately below the constraints in loser-marks. The demotion must be minimal, hence reverse ranking is suggested (Kager, 1999).

Checking Learnability of the constraints(contd.)

- **Linear stratum (for candidates in table 5): No ranking**
  $H_0 : \{ALIGN-L, ALIGN-R, CONTIGUITY, DEP-IO, DEP(MF), SONORITY\}$.
- **In the following steps, each suboptimal candidate is compared (on basis of harmony <) against the optimal candidate to deduce the loser and winner marks. The mark-data pairs are listed below-**

Mark-data pairs (subopt < opt)	Loser-marks	Winner-marks
$H_1 : e < c$	{*SONORITY,*ALIGN-L}	{*DEP-IO, *CONTIGUITY}
$H_2 : d < c$	{*DEP(MF)}	{*ALIGN-R,*DEP-IO,*CONTIGUITY}
$H_3 : b < c$	{*ALIGN-L,*DEP(MF)}	{*DEP-IO}
$H_4 : a < c$	{*SONORITY}	...

Table 6. Mark-data pairs of Table 5 candidates

- **H0 is the initial strata where the constraints remain unranked.**
- **At H1, suboptimal candidate *e*[is.piːd] is less harmonious than the optimal *c*. *DEP-IO and *CONTIGUITY are demoted immediately below H0. Thus, the winner-marks constraints are demoted to a stratum immediately below the loser-marks constraints.**
  {ALIGN-L, ALIGN-R, CONTIGUITY, DEP-IO, DEP(MF), SONORITY}
  »
  {DEP-IO,CONTIGUITY}
- **At H2, *d* [səpiːd] is compared against the optimal candidate *c* [sɨpiːd]. The table shows that Kashmiri would rather allow right-edge insertion than ungrammatical vowel insertion.**
  {ALIGN-L, DEP(MF), SONORITY}
  »
  {ALIGN-R, DEP-IO, CONTIGUITY}
- **ALIGN-R already dominates DEP-IO and CONTIGUITY in H1 demotion, hence-**
  {ALIGN-L, DEP(MF), SONORITY}
  »
  {ALIGN-R}
  »
  {DEP-IO, CONTIGUITY}
- **Minimal demotion occurs in H3 and H4 as the undominated constraints are already highest-ranked.**
- **The algorithm appears to converge into the constraint ranking of the target grammar.**

5. Discussion and future research

- Despite the fact that voiceless unaspirated stops are aspirated in the coda position, words like [šuip], [aksar], [kašt] exist in native grammar.
  - **Is there an underlying representation?**
- Gouskova (2002) proposed that if a /s/ is followed by a consonant, it cannot be broken up, but it is very regular in Kashmiri. The constraints are strictly motivated by S.S.P.
- However, there are examples like *clerk* >klərik or *park* > parikh where the cluster does follow S.S.P. yet is broken up.
  - **Are there other motivations for epenthesis?**
- Though the learner appears to converge into target grammar, the assumption that a real-life learner has access to underlying representation and acoustic cues is arbitrary.
- Tesar & Smolensky's (1993) algorithm is proved to converge even if any one of the constraints leading to the actual ranking of the dataset, the next intention of this research is to analyze the hidden structures if any.
  - **What happens when the learner is presented with incorrect input**
  - **Can it achieve the correct ranking given the hidden structure?**
  - **Can Maximum Entropy grammar or Stochastic OT account for these instances?.**

Selected References

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